

Exercise 1: Write a script that declares variables of each type (int, float, str, bool), performs arithmetic operations, and prints formatted output using f-strings.

Double-click (or enter) to edit

```
name = "Ana Maria"           #str
age = 35                     #int
height = 1.75                #float
is_student = True            #bool

result = age + 10

print(f"Student name is {name}")
print(f"Age in 10 years will be {result}")
print(f"Her height is {height}")
print(f"Is she a student? {is_student}")
```

```
Student name is Ana Maria
Age in 10 years will be 45
Her height is 1.75
Is she a student? True
```

Explanation: I have created a variable called **Ana Maria** and stored it in **name**, because its data type is text, this variable is a string. The second variable created is **35**, stored in **age**, and because it is a number, this makes it an **integer**. Third variable contains a decimal point **1.75**, and this is called a **float**. Because the last variable stores **True**, it is a **boolean**. A boolean is used when somebody has only two options, true or false. In the next line, I am asking for the **result** between **age**, after adding 10, so Python is checking what value does age contain **age + 10**. The **print** result is printing the information on the screen, the **f** before the quotation marks creates the **f-string**, inside {} Python inserts the value that is stored in the variable.

So the Output is: Student name is Ana Maria Age in 10 years will be 45 Her height is 1.75 Is she a student? Yes

Exercise 2: Implement a function that accepts two numbers and returns their sum, difference, product, and quotient as a tuple — call it with at least three different input pairs.

Double-click (or enter) to edit

```
def calculate(parameter1, parameter2):
    sum_result = parameter1 + parameter2
    difference = parameter1 - parameter2
    product = parameter1 * parameter2
    division = parameter1 / parameter2

    return sum_result, difference, product, division

result1 = calculate(5, 10)
result2 = calculate(9, 8)
result3 = calculate(7, 4)
result4 = calculate(3, 2)

print(result1)
print(result2)
print(result3)
print(result4)
```

```
(15, -5, 50, 0.5)
(17, 1, 72, 1.125)
(11, 3, 28, 1.75)
(5, 1, 6, 1.5)
```

Explanation: I've defined a function called **calculate** and it accepts two parameters **parameter1** and **parameter2**. I've added different types of calculation **++, -, *, /****. **return** statement is sending the calculated value to wherever the function was called. The program automatically groups the values into a tuple. Finally I called the function using different numbers and the result is printed into a **tuple**.

Output: (15, -5, 50, 0.5) (17, 1, 72, 1.125) (11, 3, 28, 1.75) (5, 1, 6, 1.5)

Exercise 3: Use a for loop with range() to generate a multiplication table (1–10) and print it in a formatted grid.

```
for item in range(1, 11):
```

```

for number in range(1, 11):
    print(f"{item * number:4}", end=" ")
    print()

```

```

1  2  3  4  5  6  7  8  9 10
2  4  6  8 10 12 14 16 18 20
3  6  9 12 15 18 21 24 27 30
4  8 12 16 20 24 28 32 36 40
5 10 15 20 25 30 35 40 45 50
6 12 18 24 30 36 42 48 54 60
7 14 21 28 35 42 49 56 63 70
8 16 24 32 40 48 56 64 72 80
9 18 27 36 45 54 63 72 81 90
10 20 30 40 50 60 70 80 90 100

```

Explanation: **for** means to repeat this block of code, **item** is the loop variable, **range(1, 11)** generates the number (1 to 10). The **print** line is printing the multiplication result. The **f-string** allows me to place a variable or an expression inside **{}**. **item * number:4** Python evaluates the expression inside the brackets and number 4 is telling it to print the number using at least 4 character space.

Exercise 4: Write a temperature converter that reads a value and unit (C/F) and uses if/elif/else to convert and display the result.

```

temperature = float(input("Please enter the temperature: "))
unit = input("Please enter the unit (C/F): ")

if unit == "C":
    fahrenheit = (temperature * 9 / 5) + 32
    print(f"{temperature}°C is equal to {fahrenheit}°F")

elif unit == "F":
    celsius = (temperature - 32) * 5 / 9
    print(f"{temperature}°F is equal to {celsius:.2f}°C")

else:
    print("Invalid unit. Please enter C or F.")

```

```

Please enter the temperature: 35
Please enter the unit (C/F): C
35.0°C is equal to 95.0°F

```

Explanation: First line is asking the user to input a temperature and waits for the user to type a value. The line **unit == "C"** it is checking if the value of the unit is exactly C, if yes, Python executes the block below. **print()** displays the result, and the f before the string allows Python to insert the values of the variables temperature and fahrenheit into the message. **elif** means "else if"; if the first if condition is false, Python checks whether unit is equal to "F". **else** runs if neither the if nor the elif conditions are true, meaning the user entered an invalid unit.